

Adversarial attacks for speaker recognition system with FGSM-CNN and FGSM-DNN

Mokdad Yacine Abdelhakim

Speech communication and signal processing laboratory
USTHB University
Algiers, Algeria
ymokdad@usthb.dz

Krobba Ahmed, Debyeche Mohamed

Speech communication and signal processing laboratory
USTHB University
Algiers, Algeria
akrobba@usthb.dz, mdebyeche@usthb.dz

Abstract - Adversarial attacks are specifically designed to interfere with models of speaker recognition systems by introducing imperceptible noise into speech signals. This paper explores deep learning-based adversarial attacks targeting speaker recognition systems. We used the x-vector framework and various variants of the FGSM technique applied to convolutional neural networks (CNNs) and deep neural networks (DNNs). We evaluated five attack methods: FGSM, BIM, FGSM-CNN and FGSM-DNN. Performance evaluations of the proposed methods were conducted using the LibriSpeech corpus, focusing on the equalised error rate (EER) measured by two metrics: PLDA and Cosine. The experimental results indicate that the proposed adversarial attack methods offer superior performance in terms of EER compared with the traditional FGSM approach.

Keywords - Adversarial attacks, speaker recognition, FGSM, CNN, DNN, IBM, EER, PLDA, Cosine.

I. INTRODUCTION

Speaker recognition consists of identifying a person on the basis of their voice. It is one of the most successful biometric authentication technologies, used for banking transactions, remote payments and criminal investigations [1] [2].

However, the security of speaker recognition systems is a major concern, as various attack techniques have been developed to weaken them. These attacks are generally classified into three categories: impersonation attacks, which aim to fool the speaker recognition system by presenting false voices.

There are four main subtypes: replay attacks, which use recorded voice samples to impersonate the legitimate speaker; speech conversion attacks, which modify one person's voice to sound like another's; impersonation attacks, which impersonate the target speaker's voice; and text-to-speech, which uses text-to-speech technology to create a synthetic voice that mimics the target speaker's voice.

The ASVspoof series of challenges is a significant community initiative aimed at combating these identity theft attacks. Identity theft detection methods and countermeasures to combat these attacks have developed rapidly.

Backdoor attacks: These involve introducing malicious data into the training dataset of a speaker recognition model. During the test phase, the attack is activated by the presentation of a small, specific trigger pattern, leading the system to incorrectly identify the speaker [3][4][5][6]. Adversarial attacks are designed to fool speaker recognition systems by adding subtle perturbations to benign test samples. These perturbations are often imperceptible to humans, but can significantly compromise the accuracy of deep learning-based speaker recognition systems, resulting in incorrect identification [9] [10] [11].

Adversarial attacks against machine learning systems have been the subject of extensive research in the field of image classification, leading to the proposal of various methods. Notable techniques include the Fast Gradient Sign Method (FGSM), the Basic Iterative Method (BIM) and the Jacobian-based Salinity Map Attack (JSMA) [12] [13]. Conversely, adversarial attacks against speaker recognition systems have only recently begun to attract attention, as indicated by a recent study. These attacks pose a unique challenge as they aim to fool voice-based biometric authentication systems by introducing subtle perturbations that can compromise the accuracy of the system [14] [15].

The contributions of our paper can be summarised as follows: We examine the security flaws present in three types of authentication systems: FGSM-CNN, FGSM-DNN and FGSM. In addition, we examine the mechanisms of the PLDA and Cosine metrics.

In addition, we explore attacks generated by deep learning, focusing specifically on CNN and DNN.

The remainder of the paper is structured as follows. Section II briefly reviews the work on adversarial attacks. Section III presents the FGSM function. Results and discussion are presented in Section IV. Conclusions and directions for future work are presented in Section V.

II. ADVERSARIAL ATTACKS

Adversarial attacks have been the subject of extensive research in the field of computer vision. The basic concept of these attacks can be briefly illustrated. For example, an audio

recording labelled 1 can be correctly identified by the model under normal conditions. However, the introduction of a malicious perturbation into this recording can lead to a recognition error. Most studies have focused on gradient-based perturbations in a continuous input space using a white box.

In response to criticism and to move towards a more realistic adversarial scenario, many researchers have begun to explore attack methods under black-box conditions.

In this section, we provide a brief overview of attack concepts, as well as an explanation of the ASR and speaker recognition systems to facilitate understanding. The majority of attacks in the audio domain are classified as evasion attacks. The basic idea is to introduce a disturbance into the original sound before introducing it into the ASR system. Given the unique characteristics of audio, it is generally accepted that the objective of attacking ASR systems has two key aspects: to trick the system into generating incorrect results and to avoid detection by humans.

We study current attack models and argue that achieving the first objective relies on the use of audio adversaries that introduce small perturbations into the input audio data [12][13].

Secondly, we apply the principles of psychoacoustics to ensure that the attack remains undetectable and silent. Figure 1 illustrates the overall process of an adversarial attack in voice processing systems. When the original audio is processed by the target system, it produces the correct result A.

However, when the attacker introduces a perturbation into the original audio, the target system generates an incorrect result B, which differs from A, indicating the effectiveness of the attack [14][15].

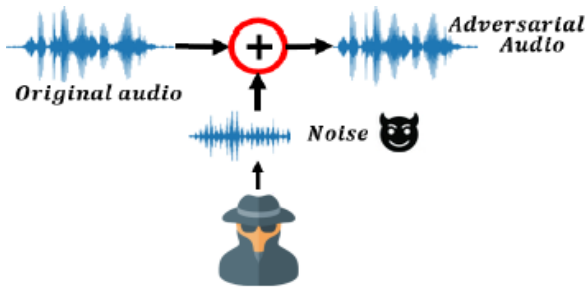


Fig. 1: Overview of the mechanism behind a malicious attack on a voice processing system.

We will therefore describe the main methods for generating adversarial examples, discuss the application of acoustic masking and present the criteria for evaluating attack models [16].

Direct attacks : Unlike attack transfer methods, direct attacks target the model itself. Many techniques exist for generating adversarial examples targeting deep learning models [17].

In order to carry out a systematic analysis of methods for generating adversarial examples, we will examine different approaches in this field [7].

- The Fast Gradient Sign Method (FGSM) is a widely used technique for rapidly creating adversarial examples. It operates by computing the gradient of the model's loss function with respect to the input data. Subsequently, the input data is adjusted slightly in the direction of the gradient to increase the loss and deceive the model [18].
- BIM (basic iterative method): Although the FGSM method is simple and computationally efficient compared to other techniques, it tends to have a lower success rate when applied to non-linear models. This is because in linear models, the direction of loss reduction is obvious and, even with multiple iterations, the direction of disturbance remains unchanged. On the other hand, for non-linear models, relying on a single iteration may not give the most accurate direction for disturbance adjustments. BIM improves on the FGSM approach by decomposing a single step into several smaller steps, allowing iterative generation of conflicting examples.
- PGD (Projected Gradient Descent) is an enhancement of the FGSM algorithm designed to create more resilient adversarial examples that are harder to detect. Rather than applying a single perturbation in the direction of the gradient, the PGD algorithm executes multiple iterations, constraining the magnitude of the perturbation at each step. This iterative approach enables a broader exploration of the search space, resulting in the generation of more effective adversarial examples [18].

III. THE FGSM FUNCTION

Initially, the FGSM was utilised to identify the minimum disturbance required for each individual input data point, sufficient to alter the model's decision [19]. The FGSM function is central to the FGSM_CNN attack process. It takes as inputs the model, the original audio, the target label, and epsilon, and outputs the modified audio. Formally, the FGSM is defined by the following equation:

$$\eta = \epsilon \text{ sign}(\nabla_x J(\theta, X, y)) \quad (1)$$

Here, $J(\theta, x, y)$ represents the gradient of the cost function concerning the model's input, output, and weights, defined as follows:

- θ : the model parameters,
- x : the model input,
- y : the model output.

Next, the sign of each gradient term is considered, resulting in a matrix of -1, 0, and 1. The perturbation matrix is then scaled by a small value, ϵ . Increasing ϵ raises the likelihood of the network making an incorrect prediction.

The resulting parameter η represents the adverse perturbations. The adversarial example x' generated by applying the FGSM method to the audio x can be calculated as follows:

$$x' = x + \eta \quad (2)$$

A. Proposed Approach

Our attack strategy is grounded in the Fast Gradient Sign Method (FGSM). While the FGSM algorithm is straightforward and efficient, it can produce adversarial instances that are readily identifiable by more advanced countermeasures [18]. Numerous variants of the FGSM have been introduced in the literature, such as MI-FGSM and NI-FGSM [21]. Therefore, our contribution aims to develop a new variant by incorporating a neural network to enhance the FGSM's predictions.

Attack algorithm :

Model prediction: This is the class to which the model predicts the audio belongs, which is then used to perform the calculations to track the gradients. This prediction is generated by a convolutional neural network (CNN).

Loss calculation: The loss function is determined by comparing the model's prediction with the target label, by measuring the difference between the predicted label and the correct label.

Gradient calculation : The gradient indicates how the audio input should be adjusted to minimise loss.

Creation of a contradictory disturbance: The adverse disturbance is generated by multiplying the sign of the gradient by epsilon, which controls the strength of the modification. The sign of the gradient indicates the direction in which the audio should be modified.

Adding the perturbation to the original audio: The perturbation is then added to the original audio, resulting in a new audio sample that is slightly modified but likely to fool the model into predicting the target label.

B. Use of the loss function in the attack

Carlini and Wagner have shown that the selection of the loss function can significantly affect the final distortion of generated adversarial examples, potentially by a factor of three or more [8]. The aim of the attack is to create a perturbed sound, which involves modifying the sound in such a way as to increase the probability that the model predicts an incorrect target label.

The loss function is used to quantify the difference between the model's predictions and the actual labels. By minimising this loss, we can train the model to make accurate classifications. For classification tasks, the "binary cross-entropy" function is generally used. The attack uses this loss function to calculate the gradient for the input audio.

The gradient is derived with respect to the model parameters and indicates the direction in which these parameters should be adjusted to reduce the loss. We apply the "Adam" optimizer to solve this update formulation. Following the direction of the gradient, the adverse perturbation is generated by multiplying the gradient by epsilon (which controls the strength of the perturbation) and adding it to the original audio.

The epsilon value is a crucial parameter that dictates the intensity of the attack. A higher epsilon value leads to greater disturbance of the audio, which makes it easier to fool the model, but can also make the audio sound less natural. We took the value $\epsilon = 0.002$, for both types of CNN and DNN networks.

C. Neural network architecture

CNN: This model has five layers: 1D convolution layer, 1D Max Pooling layer, Flatten layer, and two Dense layers.

DNN: We use four layers, three of which are Dense layers;

ReLU and Sigmoid were used as activation functions for both networks.

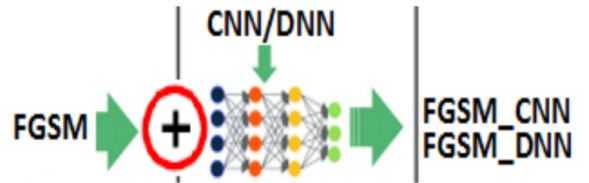


Fig. 2: Process of adapting the FGSM to the CNN and DNN models.

IV. RESULTS AND DISCUSSION

In this section, we describe the experiments carried out to evaluate the attack framework. Section 4.1 details the dataset and model parameters, section 4.2 examines the attack parameters and section 4.4 presents the experimental results.

A. Experimental set-up

Our experimental setup is based on the selection of the speaker's audio recording database, the model used and the evaluation measures chosen for the results. We used the LibriSpeech open access speech dataset, which comprises approximately 1,000 hours of English speech from audiobooks read as part of the LibriVox project. The data is meticulously

segmented and aligned. For our experiments, we selected a 100-hour subset of the training dataset, as well as the clean development set and the clean test set.

These specific types of speech sets are often needed for various recognition tasks, such as those requiring recorded speakers for speaker verification (SV) and open set identification (OSI) tasks.

In all experiments, we used audio recordings encoded in "flac" format with mono channels and a sampling rate of 16 kHz [7].

As far as the model is concerned, we opted for the x-vector model for the OSI task; this system represents an advance on the i-vectors originally designed for speaker verification. Prototypical speaker embeddings, known as i-vectors, are derived from Gaussian-mixture-model-based universal background models. Recently, with the rapid advancement of deep learning, deep speaker embeddings have emerged as a prominent trend. Notable examples of these deep embeddings include the x-vector [22].

The x-vector deep learning approach has achieved state-of-the-art performance in many scenarios and applications. After x-vector extraction, speaker classification is performed using various methods, including PLDA and Cosine Similarity.

B. Evaluation parameters

To evaluate the effectiveness of our attack, we generate targeted adversarial audio samples based on 2620 instances of the test-clean database, which includes 40 speakers. For each speaker, we target all existing audio samples. We focus primarily on the degree of evaluation, the costs of the attack, which are measured by the Equal Error Rate (EER), which is generated for each adversarial example.

TABLE 1. ACCURACY AND PERFORMANCE OF DIFFERENT ETCHING METHODS FOR PLDA AND COSINE MEASUREMENTS

		Methods			
Metric		FGSM	FGSM_CNN	FGSM_DNN	BIM
EER%	PLDA	5.31	5.025	6.79	13.14
	Cosine	8.03	7.46	9.63	14.40
Accuracy (%)		97.92	94.45	97.92	97.92

C. Assessment of attacks

In order to evaluate the performance of the proposed methods in detail, we compared the experimental results of our methods with the results of a reference method such as FGSM and BIM, as shown in Table 1.

Five methods were tested, and performance was assessed in terms of accuracy and equalised error rate (EER) using two metrics: PLDA and Cosine.

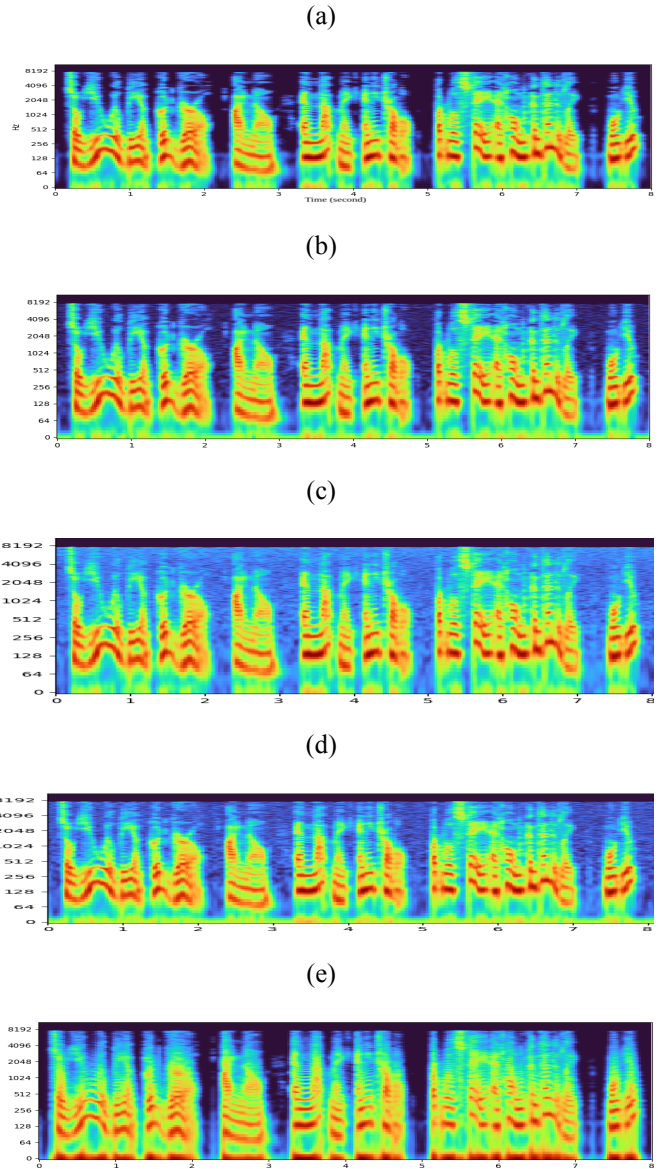


Fig. 3 Spectrogramme of speech for an utterance «r», (a) Clean, (b) FGSM, (c) FGSM-CNN, (d) FGSM-DNN, (e) BIM.

D. Analysis of results

The results show a significant difference in accuracy and EER performance between the different methods.

FGSM_CNN has a slightly lower accuracy (94.45%) than FGSM_DNN and FGSM (both 97.92%). However, it has a better EER using PLDA (5.02) than FGSM_DNN (6.79) and FGSM (5.31).

FGSM_DNN, although having the same accuracy as FGSM, has higher EERs for both metrics (PLDA and Cosine).

This suggests that, despite comparable accuracy, FGSM_DNN is less robust against gradual perturbation attacks in terms of error rate.

FGSM shows an intermediate performance with EERs lower than those of FGSM_DNN but higher than those of FGSM_CNN using the PLDA metric. For the Cosine metric, FGSM (8.03) is also better than FGSM_DNN (9.63) but underperforms FGSM_CNN (7.46).

Concerning BIM, these values PLDA and Cosine, are very high compared to other methods.

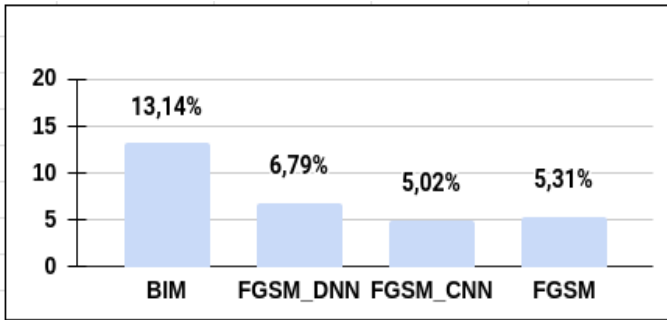


Fig. 4 Performance with EER PLDA

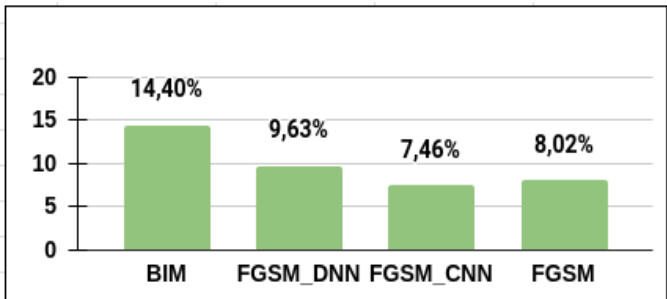


Fig. 5 Performance with EER Cosine

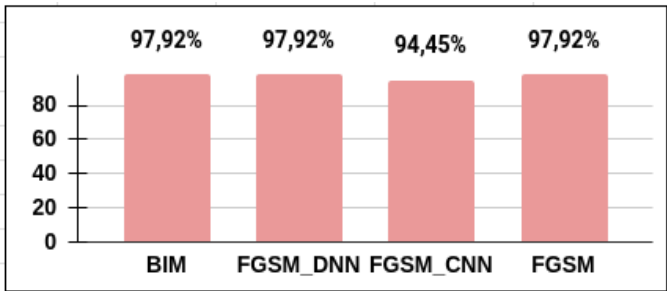


Fig. 6 Performance with Accuracy

These results indicate that the structure of the attack model (CNN vs. DNN) and the method used (FGSM) significantly influence the performance of speaker recognition attacks. In general, CNN-based models appear to offer better resistance to adverse perturbations than DNN-based models, as shown by the lower EERs. This study highlights differences

in the robustness of speaker recognition systems to adversarial attacks, which vary according to the model architectures and attack methods employed. CNN models show better resistance than DNN models, as shown by the lower EERs. Future research should focus on improving the robustness of systems by exploring combinations of different model architectures and attack techniques to provide more reliable voice security.

V. CONCLUSION

In this paper, we examined adversarial attacks in the context of the security of speaker recognition (SR) systems in speech processing systems. We started by giving an overview of speaker recognition systems and their tasks, as well as existing attack methods and the psychoacoustic model. Next, we presented a threat model that includes two new adversarial attacks based on FGSM, CNN and DNN. Finally, we evaluated the experimental results of these attacks and compared their performance with other methods such as BIM, using the equal error rate (EER) of the x-vector model as a metric. We anticipate that future research will continue to explore adversarial attacks on voice processing systems, with a focus on adversarial attacks in speaker recognition systems, integrating a practical context and implications for industry could be as follows:

Security risks: bank authentication or voice assistants: these flaws could allow attackers to access sensitive information.

Economic impact: Companies could avoid financial, legal and reputational losses by strengthening the security of their systems against these attacks.

By integrating these elements, our research can help professionals to strengthen the security of voice recognition systems and better prepare for the real challenges posed by hostile attacks in this sector.

This research study aims to provide developers with an in-depth understanding of the characteristics of FGSMs, thereby facilitating the development of robust defence strategies against adversarial threats in deep learning models.

REFERENCES

- [1] D. A. Reynolds, R. C. Rose, "Robust text-independent speaker identification using Gaussian mixture speaker models," IEEE Trans. Speech Audio Process, vol. 3, no. 1, pp. 72-83, Jan. 1995.
- [2] D. A. Reynolds, T. F. Quatieri, R. B. Dunn, "Speaker verification using adapted Gaussian mixture models," Digit. Signal Process, vol. 10, no. 1/3, pp. 19-41, 2000.
- [3] Z. Wu, et al, "Asvspoof: The automatic speaker verification spoofing and countermeasures challenge," IEEE J. Sel. Topics Signal Process, vol. 11, no. 4, pp. 588-604, Jun. 2017.
- [4] L. Zhang, Y. Meng, J. Yu, C. Xiang, B. Falk, and H. Zhu, "Voiceprint mimicry attack towards speaker verification system in smart home," in Proc. IEEE INFOCOM Conf. Comput. Commun, pp. 377-386, 2020.
- [5] G. Chen et al, "Who is real Bob Adversarial attacks on speaker recognition systems", Proc. IEEE Symp. Secur. Privacy, pp. 694-711, 2021.

- [6] A. Ilyas, L. Engstrom, A. Athalye, J. Lin, "Black-box adversarial attacks with limited queries and information", *Proc. Int. Conf. Mach. Learn.*, pp. 2137-2146, 2018.
- [7] Z. Sun, Y. Ren, Y. Huang, W. Liu, H. Zhu, "AFPM: A Low-Cost and Universal Adversarial Defense for Speaker Recognition Systems", *IEEE Transactions on Information Forensics and Security*, 2023.
- [8] X. Zhang, Y. Xu, S. Zhang, X. Li, "A highly stealthy adaptive decay attack against speaker recognition", *IEEE Access*, 10, 118789-118805, 2022.
- [9] H. Wu, X. Li, A. T. Liu, Z. Wu, H. Meng, H. Y. Lee, "Improving the adversarial robustness for speaker verification by self-supervised learning", *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 30, 202-217, 2021.
- [10] X. Wang, Z. Yan, R. Zhang, P. Zhang, "Attacks and defenses in user authentication systems: A survey", *Journal of Network and Computer Applications*, 188, 103080, 2021.
- [11] X. Chen, S. Li, H. Huang, "Adversarial Attack and Defense on Deep Neural Network-Based Voice Processing Systems: An Overview", *Appl. Sci*, 11, 8450, 2021.
- [12] A. Z. Zaidi, A. S. Sadiq, "Touch-based continuous mobile device authentication: State-of-the-art, challenges and opportunities", *Journal of Network and Computer Applications*, 2021.
- [13] Y. Zhang, J. Yang, X. Li, H. Liu, K. Shao, "Textual Adversarial Attacking with Limited Queries", *Electronics* 10, 2671, 2021.
- [14] N. Carlini, D. Wagner, "Audio Adversarial Examples: Targeted Attacks on Speech-to-Text", University of California, Berkeley.
- [15] <https://www.mathworks.com/help/audio/ug/speaker-recognition-using-x-vector>.
- [16] R. K. Das, X. Tian, T. Kinnunen, H. Li, "The attacker's perspective on automatic speaker verification: An overview", *arXiv:2004.08849*, 2020.
- [17] M. Marras, P. Korus, A. Jain, N. Memon, "Dictionary Attacks on Speaker Verification", *IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY*, VOL. 18, 2023.
- [18] W. Villegas-Ch, A. Jaramillo-Alcázar, S. Luján-Mora, "Evaluating the Robustness of Deep Learning Models against Adversarial Attacks: An Analysis with FGSM, PGD and CW" *Big Data Cogn. Comput.*, 8, 8, 2024.
- [19] N. Ding, K. Möller, "Minimally Distorted Adversarial Images with a Step-Adaptive Iterative Fast Gradient Sign Method", *AI*, 5, 922-937, 2024.
- [20] H. ISSAOUI, A. ELADEL, A. ZOUINKHI, M. ZAIED, L. KHRIJI, S. H. NENGROO, "Defending CNN Against FGSM Attacks", *IEEE Access*, VOLUME 12, 2024.
- [21] D. Hong, D. Chen, Y. Zhang, H. Zhou, L. Xie, "Attacking Robot Vision Models Efficiently Based on Improved Fast Gradient Sign Method", *Appl. Sci*, 14, 1257, 2024.
- [22] J. Yao, H. Luo, J. Qi, X. Zhang, "Interpretable Spectrum Transformation Attacks to Speaker Recognition Systems", *IEEE/ACM TRANSACTIONS ON AUDIO, SPEECH, AND LANGUAGE PROCESSING*, VOL. 32, 2024.